Brief Documentation — Automation & QA Assessment

Nikhil · Repo: github.com/Kryptonnnnnn/automation-qa-assignment

Deliverables: `Task1_QA_Report_Nikhil.pdf` · `Task2_Workflow_Nikhil.json` · `Bonus_UptimeMonitor.json` · `screenshot/` · `README.md` (setup details)

Task 1 — QA & Debug Report

- **App:** TodoMVC (React), todomvc.com/examples/react/dist. The RealWorld demo and API are offline (404 / Cloudflare 1016), so I used TodoMVC.
- **How I tested:** every main flow (add, complete, edit, delete, filter, clear, refresh), plus empty, long and HTML input, keyboard-only use, touch and 320 px mobile. I ran it in headless Chromium (Playwright) and inspected the shipped JS bundle.
- **Result:** 8 issues (2 High, 3 Medium, 3 Low) in the required table. The biggest: todos are lost on refresh, and the delete button is unusable on touch and by keyboard.
- **Root-cause analysis:** data loss on refresh. State lives only in React memory and the bundle contains no `localStorage` code. Fix: initialise state from `localStorage` and write it back on every change.

Task 2 — n8n “Morning Brief” workflow

Every hour → CoinGecko markets → Top 5 gainers (Code) → CoinGecko 24h chart (enrichment) → Build digest → IF top gainer > 5% → Discord (Hot Alert / Normal Digest)

- **Why these APIs:** CoinGecko is free and needs no key, and its data changes hourly. Discord webhooks are free and simple.
- **Transformation:** drop coins with no 24h change, sort by 24h % descending, keep the top 5.
- **Threshold:** 5% is a notable daily move for large-cap coins, so those go out as a “Hot move” alert.
- **Error handling:** HTTP nodes retry 3 times, then use an error output → Format Error → Discord failure alert. If only the chart call fails, the digest is still sent without it.
- **Credentials:** a Discord Webhook credential in n8n; no secrets in the exported JSON.

Bonus — Uptime monitor

- Pings the app every 5 minutes and checks the HTTP **status code**. A non-200 is re-checked after 15 s, and only a repeated failure sends a Discord alert (avoids false alarms).
- Tracks response time. A daily 09:00 summary reports availability and average / slowest response time.

Verification

- Both workflows were run in a local n8n (2.22.6) against the live CoinGecko API. The normal run, the broken-API error alert and the app-down alert all behaved as designed. Screenshots are in `screenshot/`.